

AL BURAIMI, Oman

WMO: 412440

Lat: 24.2337N

Lon: 55.9155E

Elev: 372

StdP: 96.94

Time Zone: 4.00 (E04)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.6	11.9	-6.0	2.4	28.2	-4.0	2.8	30.1	9.6	22.7	8.3	21.5	1.1	130	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	44.9	20.8	44.0	20.9	43.2	21.0	27.1	33.3	26.3	34.0	25.5	34.6	4.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	21.8	30.8	24.3	20.1	31.4	22.9	18.4	31.5	88.2	33.3	84.5	33.9	80.9	34.5	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.7	6.7	DB	7.9	46.4	1.9	0.6	6.5	46.9	5.4	47.2	4.3	47.5	2.9	48.0	(4)
(5)			WB	4.6	28.3	1.6	0.8	3.5	28.9	2.5	29.4	1.7	29.8	0.5	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	18.4	20.4	24.0	28.9	33.4	36.0	37.1	37.1	34.3	30.1	24.3	20.1	(6)	
	DBStd	7.13	2.40	3.12	3.32	3.15	2.37	2.04	2.03	1.55	1.77	1.85	2.30	2.28	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	53	29	13	2	0	0	0	0	0	0	0	0	10	(9)	
	CDD10.0	6832	261	290	433	567	727	781	841	839	728	622	430	314	(10)	
	CDD18.3	3843	31	69	176	317	468	531	582	581	478	364	180	66	(11)	
	CDH23.3	58869	183	597	1941	4333	7533	9044	10108	10109	7793	5097	1731	401	(12)	
	CDH26.7	39825	12	157	868	2558	5207	6665	7633	7630	5417	2977	649	51	(13)	
(14) Wind	WSAvg	3.1	2.8	3.1	3.3	3.3	3.2	3.4	3.5	3.5	3.3	3.0	2.7	2.6	(14)	
(15) Precipitation	PrecAvg	90	11	17	21	11	1	3	7	5	3	1	2	8	(15)	
	PrecMax	263	54	77	86	62	4	29	101	14	35	9	18	52	(16)	
	PrecMin	18	0	2	0	0	0	0	0	3	0	0	0	0	(17)	
	PrecStd	58	14	18	27	14	1	6	19	3	6	3	5	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.2	32.1	36.2	40.4	44.6	46.1	45.8	45.4	43.6	39.7	34.2	29.5	(19)	
		MCWB	15.7	16.4	16.4	18.5	19.3	20.8	21.1	21.1	20.5	19.0	17.9	16.6	(20)	
	2%	DB	26.5	30.2	34.5	38.9	42.7	44.7	44.8	44.3	42.2	38.3	32.8	28.1	(21)	
		MCWB	15.6	16.0	16.3	18.3	19.3	20.3	21.2	21.3	20.3	18.8	17.8	16.5	(22)	
	5%	DB	25.4	28.6	33.1	37.6	41.5	43.7	43.9	43.3	41.1	37.2	31.6	26.9	(23)	
		MCWB	15.2	15.7	16.2	17.9	19.4	20.3	21.4	21.4	20.4	18.7	17.7	16.4	(24)	
	10%	DB	24.2	27.0	31.5	36.2	40.4	42.7	42.8	42.4	40.1	36.0	30.4	25.7	(25)	
		MCWB	14.6	15.4	15.9	17.6	19.3	20.4	21.7	21.6	20.6	18.5	17.6	16.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	19.7	21.1	22.1	24.2	27.1	28.1	28.0	27.1	25.0	22.2	19.9	(27)	
		MCDB	21.8	26.6	27.6	32.0	32.6	32.9	32.4	33.4	31.5	29.3	26.2	23.8	(28)	
	2%	WB	17.7	18.7	19.9	20.8	23.1	25.7	27.1	26.9	26.0	23.7	21.2	18.9	(29)	
		MCDB	22.4	24.8	27.1	31.4	33.8	33.8	33.3	34.2	31.4	29.0	26.1	23.9	(30)	
	5%	WB	16.8	17.9	18.8	19.9	22.0	24.7	26.4	26.0	25.0	23.0	20.4	18.2	(31)	
		MCDB	22.6	24.2	27.1	31.2	34.7	34.2	34.4	35.4	32.2	28.9	26.1	23.3	(32)	
	10%	WB	16.0	17.0	17.9	19.2	21.0	23.9	25.5	25.0	24.1	22.2	19.5	17.6	(33)	
		MCDB	22.2	23.9	26.1	30.7	34.9	34.6	35.3	36.2	33.0	29.6	26.1	23.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.5	12.5	12.6	13.5	13.4	11.6	11.3	11.9	12.0	11.0	10.3	(35)	
		MCDBR	11.6	12.9	14.0	13.9	14.6	14.1	12.4	12.1	12.5	12.9	12.0	11.5	(36)	
	5% WB	MCWBR	4.6	5.0	4.9	4.7	4.8	4.7	4.5	4.5	5.2	5.5	4.9	4.6	(37)	
		MCWBR	9.5	10.0	10.3	11.4	13.1	13.3	11.9	11.8	12.4	12.5	11.4	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.413	0.466	0.529	0.569	0.571	0.635	0.720	0.686	0.583	0.480	0.427	0.408	(40)	
		taud	2.112	1.948	1.786	1.728	1.721	1.620	1.519	1.572	1.730	1.940	2.096	2.125	(41)	
	Ebn at Noon		831	811	780	758	753	701	644	665	725	786	809	818	(42)	
		Edh at Noon	141	178	220	238	239	262	290	274	229	177	142	135	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.38	5.15	6.02	6.77	7.51	7.57	7.01	6.82	6.42	5.69	4.67	4.16	(44)	
	RadStd		0.24	0.32	0.36	0.41	0.19	0.21	0.25	0.17	0.12	0.12	0.10	0.21	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.47	+0.70	N/A	N/A	N/A	N/A	N/A	+150	+149

Nomenclature: See separate page